Surveillance Pipeline Architecture & Technical Verification

1. End-to-End Processing Architecture

Stage 1: Spatial Gating & Motion Analysis: Evaluates native frames within the calibrated mechanical boundary [295, 20, 1490, 780] to filter irrelevant terrace activity.
Stage 2: Lossless Bitstream Extraction: Executes direct FFmpeg HEVC packet copying with enforced ±60-second pre- and post-event contextual buffers.
Stage 3: Deep Feature Extraction: Crops faces passing confidence >= 0.80 and quality score Q >= 40.0, extracting L2-normalized 512-D ArcFace vectors.
Stage 4: Ground-Truth Biometric Graph Consolidation: Groups vectors via batched ArcFace pairwise similarity (>= 0.55) and graph connected component analysis to establish true human identities.

2. Root Cause Analysis: Unsupervised Clustering vs Graph Consolidation

• **The Hyper-Fragmentation Problem:** Unsupervised DBSCAN with a strict distance threshold (d <= 0.38) shatters identical human faces across different lighting conditions, head angles, and days into hundreds of micro-clusters. Over 87% of initial clusters contained only 2–4 detections.
• **The Graph Consolidation Solution:** By computing pairwise cosine similarity across high-frequency exemplars and applying graph connected components at >= 0.55, fragmented clusters mathematically merged into the true plant population of 43 unique individuals.
• **Direct Incident Face Verification:** Rather than inferring identity from temporal proximity, every incident was verified against direct face crops captured during the exact event window.

3. Calibrated Native 3K Spatial Geometry & Equipment Zones

To ensure exact bounding box alignment between raw video files and NVR monitoring stations, coordinates are mapped directly onto the native 3K (2880 x 1620) plane:

Equipment Hazard Zone	Native 3K Coordinates [X1, Y1, X2, Y2]	Dimensions	Criticality Level
Stage 1 Observation Perimeter	**[295, 20, 1490, 780]**	1195 x 760 px	Boundary Gating Mask
Zone 1A (Skid Multiport Valve)	**[1080, 370, 1210, 500]**	130 x 130 px	**CRITICAL HAZARD**
Zone 1B (Filter Head Multiport Valves)	**[810, 440, 990, 580]**	180 x 140 px	**CRITICAL HAZARD**
Zone 2 (Ultimate Power Switch)	**[995, 340, 1095, 495]**	100 x 155 px	**SEVERE HAZARD**
Zone 3 (RO Skid Controls & Walkway)	**[1080, 320, 1420, 790]**	340 x 470 px	ELEVATED MONITORING

4. Lossless Bitstream Extraction Performance Benchmark

• Direct FFmpeg bitstream extraction (-c copy) bypasses costly CPU/GPU re-encoding and preserves 100% bit-for-bit forensic image quality.
• **Benchmark Throughput:** Successfully extracted **20 pristine MP4 incident clips (0.91 GB) in 108.7 seconds** directly from read-only storage (**>250x realtime speed**).